

Yan Lukashevich

Python Developer — backend · automation · scientific computing

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PROFESSIONAL PROFILE

Python developer with two years of commercial experience in scientific computing: I designed and wrote a computational library used by an NCU research group and automated the computational cycle on an HPC cluster, cutting a batch of calculations from ~a day to an hour. I also use Python in production outside academia — a local FastAPI agent drives the autonomous IoT kiosks of a commercial printing system (DrukMruk) that I built and deployed entirely on my own (.NET, React, Azure). I analyze and visualize data (NumPy, pandas, matplotlib), train ML models (scikit-learn), and integrate LLMs into applications.

TECH STACK

- **Python:** FastAPI, NumPy, SciPy, pandas, matplotlib, Jupyter, pytest, MicroPython · Psi4 (quantum chemistry)
- **Automation / HPC / Linux:** bash, HPC cluster (PBS, conda), network administration (AP, DHCP, DNS, NAT, SSH), Cisco CCNA
- **AI / ML / data analysis:** scikit-learn, LLM integration (chatbots / AI assistants) into applications, data analysis and visualization
- **Web (second stack):** C#, ASP.NET Core 8, React 19, TypeScript, Node.js / NestJS, REST API, WebSocket
- **Cloud / DevOps:** Microsoft Azure (App Service, SQL, Service Bus, Key Vault, VNet, Managed Identity), GitHub Actions (CI/CD, OIDC), Docker
- **Databases / testing:** Azure SQL, SQLite, MongoDB · pytest, TDD, integration testing · Git / GitHub flow, code review

WORK EXPERIENCE

DrukMruk — Founder & Lead Developer

Sole proprietorship · 02.2026 — present · [drukruk.pl](#)

- **From zero to production in 3 months, solo** — a commercial, distributed self-service printing system: a web app, a cloud backend, and a network of physical IoT kiosks driven by an agent written in Python.
- Wrote the **local kiosk agent (FastAPI)** in Python — controls printing and the touchscreen interface on a Raspberry Pi 5; the device runs unattended in a public space, on a dedicated LTE connection, hardened in layers (defense-in-depth) against the real-world threat model of a public location.
- Passed a **month-long security audit by a public institution (UCI NCU)** — defended the architecture and hardened the system; a positive assessment was a precondition for deployment in the university's infrastructure.
- Built and maintain the cloud backend with a fault-tolerant payment integration (ASP.NET Core 8, Azure, Autopay) — cryptographically verified webhooks, automatic refunds, payment–print state consistency; critical flows covered by tests (TDD); CI/CD in GitHub Actions (OIDC).

Nicolaus Copernicus University in Toruń — Scientific programmer (quantum chemistry)

Contract · 02.2024 — 03.2026

- Designed and wrote a **Python computational library** used by the entire research group — a modular architecture that speeds up calculations relative to methods of the same accuracy class and lets the scientific community extend it easily; built on the group's body of research and the literature, being prepared for open-source publication.
- Automated the computational cycle on the cluster, cutting a batch of calculations **from ~a day to an hour** — custom scripts for queuing and launching jobs and for system monitoring (assessing the library's impact on cluster resources); configured users' computing environments (Linux, conda, PBS).
- Verified the method's correctness by demonstrating agreement with the scientific literature and reference data — analysis and interpretation of numerical results (NumPy), with visualizations and comparisons that serve as proof of the approach's validity for an upcoming publication.

PROJECTS

Ventilator simulator — Team leader

2026 · [github.com/yanlukashevich/Respirator-simulator](#)

- Led a 4-person team to **1st place in the NCU Institute of Physics competition** and to laureate status in the university-wide competition — a distributed, real-time ventilator simulator; the project is entering commercialization as an **NCU spin-off company**.
- Designed the physical patient stations (Raspberry Pi, knobs) that simulate various medical conditions — based on consultations with domain experts; owned the integration of components from different authors (Node.js / NestJS, WebSocket, React, Raspberry Pi).
- Designed a network that scales the system **from 2 to 15 stations** — built from scratch (Linux, AP / DHCP / DNS / NAT / SSH).

EDUCATION

- **Applied Computer Science**, B.Eng. engineering studies — Nicolaus Copernicus University in Toruń, 2022–2026